

Intersection homology of swallowtail

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Abstract

I compute the intersection homology of $\mathbb{R}^2$ with a singular point, $\mathbb{R}^2$ with two singular lines, and swallowtail space.

Contents

1 Intersection homology	1
2 Compute $\mathbb{R}^2$ with two degeneracy lines	5
3 Compute $\mathbb{R}^2$ with one singular point	12
4 Intersection homology of swallowtail	13
5 An interesting phenomenon	20

1 Intersection homology

In recent years, intersection homology has become an indispensable tool for studying the topology of singular spaces. While the main results of usual homology theories often fail for singular spaces, intersection homology effectively recovers these properties, bridging this critical gap. Appropriate reference for intersection homology/homotopy are [1] and [2].

Definition 1.1. A filtration is a sequence of closed subsets of X :

$$X^n \supseteq X^{n-1} \supseteq \dots \supseteq X^2 \supseteq X^1 \supseteq X^0 \supseteq X^{-1}$$

The connected component of $X^i - X^{i-1}$ is called stratum.

- In application, X^i is always i -dimension singularities of X
- A stratum in $X^i - X^{i-1}$ can be view as i -dimension spaces without lower dimensional singularities.

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Let X be a filtered simplicial complex. We say i -simplex σ in general position of stratum S if $\dim(\sigma \cap S) \leq \dim(\sigma) + \dim(S) - n$ for every stratum S of X .

It is always possible for us to move an i -simplex to be in a general position with stratum S in manifolds. However, this is not true in singular spaces.

Definition 1.2. Let X be filtered space of formal dimension n . Let $\mathcal{F}$ be the set of strata of X . A perversity on X is a function $\bar{p} : \mathcal{F} \rightarrow \mathbb{Z}$ such that $\bar{p}(S) = 0$ if $S \subset X^n - X^{n-1}$.

Definition 1.3. i -simplex σ is called $\bar{p}$ -allowable if $\dim(\sigma \cap S) \leq \dim(\sigma) + \dim(S) - n + \bar{p}(S)$ for every stratum S of X .

We tolerate to what extent the strangeness of the i -simplex; if it is too strange, we do not acknowledge it as an allowed i -chain, but if it is not too strange, we accept it as an allowable i -simplex.

Definition 1.4. Let X be a filtered simplicial simplex with filtration $X_n \supset X_{n-2} \supset \cdots \supset X_0$.

An i -chain ζ is called $\bar{p}$ -allowable if every simplex in ζ and $\partial\zeta$ is $\bar{p}$ -allowable.

Define the group $I_p C_i(X)$ be the subset of $C_i(X)$ consisting of $\bar{p}$ -allowable i -chains.

It can be shown that the chain complex $(C_*(X), \partial)$ restricts to a chain complex $(I_p C_*(X), \partial)$.

Definition 1.5. The intersection homology groups are defined by $I_p H_i(X) = H_i(I_p C_*(X))$.

For a topological space, we need PL intersection homology which is independent of triangulation. Let's define PL homology first.

Definition 1.6. The simplicial complex K' is a *subdivision* of K if:

- $|K| = |K'|$
- Every simplex of K' is contained in some simplex of K

Definition 1.7. A triangulation T of a topological space X is a pair $T = (K, h)$ where

- K is a locally finite simplicial complex
- $h : |K| \rightarrow X$ is a homeomorphism

Remark 1.8. Locally finite means: For all $x \in |K|$, there exists a neighborhood U such that U intersects only finitely many simplices.

Definition 1.9. Let $T = (K, h)$ and $S = (L, j)$ be two triangulations. We say T is equivalent to S if and only if $j^{-1}h$ is a simplicial isomorphism.

Definition 1.10. A subdivision of $T = (K, h)$ is a pair $T' = (K', h)$, where K' is a subdivision of K .

Definition 1.11. A PL (piecewise linear) space is a topological space X with

$$\mathcal{T} = \{\text{locally finite triangulations}\}$$

such that:

- For every $T \in \mathcal{T}$, every subdivision of T is also in $\mathcal{T}$.
- For every $T, S \in \mathcal{T}$, T and S have a common refinement.

Remark 1.12. Let $T = (K, h)$ and $S = (L, j)$. T, S have a common refinement means: there exists a subdivision $T' = (K', k)$ of T and a subdivision $S' = (L', \ell)$ of S such that the induced map $f : K' \rightarrow L'$ is a simplicial isomorphism.

Construction 1.13. We define a relation $\leq$ on $\mathcal{T}$ as following: Let $T = (K, h), S = (L, l) \in \mathcal{T}$

Define $T \leq S \iff S$ equivalent to a subdivision of T .

Fact 1.14. $(\mathcal{T}, \leq)$ is a directed set (i.e., the relationship $\leq$ satisfying (1) transitive (2) reflexive (3) for any $T, S \in \mathcal{T}, \exists W \in \mathcal{T}$ such that $T \leq W$ and $S \leq W$)

Definition 1.15. Let $(X, \mathcal{T})$ be a PL space. For $T = (K, k) \in \mathcal{T}$, define $C_*^T(X) = C_*(|K|)$.

Definition 1.16. Let $T = (K, k) \leq T' = (K', k')$ in $\mathcal{T}$, we want to define a map $C_*^T(X) \rightarrow C_*^{T'}(X)$, called subdivision chain map as following:

$$C_*^T(X) = C_*(|K|) \longrightarrow C_*(|K'|) = C_*^{T'}(X), \quad \sigma \mapsto \sum_{\tau_\sigma \subseteq |\sigma|} \tau_\sigma$$

For $\xi = \sum_i a_i \sigma_i$, the map is

$$\xi \mapsto \sum_i a_i \sum_{\tau_{\sigma_i} \subseteq |\sigma_i|} \tau_{\sigma_i}$$

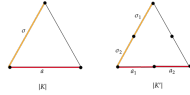


Figure 1: An example for subdivision chain map

Example 1.17.

$$\sigma \mapsto \sigma_1 + \sigma_2$$

$$\sigma + 2a \mapsto \sigma_1 + \sigma_2 + 2(a_1 + a_2)$$

It's easy to check that with above data, we can define a direct limit as following:

Definition 1.18. $C_*^{\mathcal{T}}(X) := \varinjlim_{T \in \mathcal{T}} C_*^T(X)$, where $C_*^T(X) = C_*(|K|)$ for $T = (K, k)$

Actually, we do not need all triangulations in $\mathcal{T}$. We only need a subset of $\mathcal{T}$ containing all subdivisions of a fixed triangulation T_0 .

Property 1.19. Let X be a PL space with admissible triangulation $\mathcal{T}$. Let $T_0 = (K, k) \in \mathcal{T}$ and let $\mathcal{T}_0$ be the subset of $\mathcal{T}$ consisting of subdivision of T_0 . Then $C_*^{\mathcal{T}}(X) = \varinjlim_{T \in \mathcal{T}} C_*^T(X) \cong \varinjlim_{T \in \mathcal{T}_0} C_*^T(X)$.

Definition 1.20. X is a PL space. Define PL homology of PL space $(X, \mathcal{T})$ as $\mathcal{H}_*(X) := H_*(C_{\bullet}^{\mathcal{T}}(X))$

Property 1.21. Let X be a PL space. Then $\mathcal{H}_*(X) \cong H_*(X)$, where $H_*(X)$ can be singular or simplicial homology with respect to any triangulations.

Remark 1.22. PL intersection homology (defined later) may not isomorphic to $H_*(X)$ in general!

Next, we can define PL intersection homology.

Definition 1.23. Let X be a PL filtered space such that every skeleton X^i is a subcomplex of any admissible triangulation.

Define $I^{\bar{p}}C_*^{\mathcal{T}}(X) := \varinjlim_{T \in \mathcal{T}} I^{\bar{p}}C_*^T(X)$, where $I^{\bar{p}}C_*^T(X) := I^{\bar{p}}C_*(|K|)$.

Remark 1.24. Filtration and perversity of X can “move to” $|K|$ by homeomorphism k .

Fact 1.25. If $T \leq T'$, the subdivision chain map $\bar{\nu} : C_*^T(X) \rightarrow C_*^{T'}(X)$ restricts to a map $\nu : I^{\bar{p}}C_*^T(X) \rightarrow I^{\bar{p}}C_*^{T'}(X)$

Definition 1.26. We define the PL intersection homology of PL space $(X, \mathcal{T})$ as $I^{\bar{p}}\mathcal{H}_*(X) := H_*(I^{\bar{p}}C_{\bullet}^{\mathcal{T}}(X)) \cong \varinjlim_{T \in \mathcal{T}_0} H_*(I^{\bar{p}}C_{\bullet}^T(X)) = \varinjlim_{T \in \mathcal{T}_0} I^{\bar{p}}H_*^T(X)$, where

$I^{\bar{p}}H_*^T(X) = I^{\bar{p}}H_*(|K|)$ for $T = (K, k)$.

Definition 1.27. Let L be a subcomplex of K . L is called a full subcomplex if

$$\forall \sigma \in K \text{ with vertices in } L, \text{ then } \sigma \in L$$

Definition 1.28. An admissible triangulation T of PL filtered space X with filtration $\{X^i\}$ is called full triangulation if all X^i is a full subcomplex of X .

Theorem 1.29. Let X be a PL filtered space, T a full triangulation of X , and T' any subdivision of T . Then

$$I^{\bar{p}}C_*^T(X) \longrightarrow I^{\bar{p}}C_*^{T'}(X)$$

is an isomorphism.

Corollary 1.30.

$$I^{\bar{p}}\mathcal{H}_*(X) = \varinjlim_{T' \in \mathcal{T}_0} H_*(I^{\bar{p}}C_{\bullet}^{T'}(X)) = H_*\left(\varinjlim_{T' \in \mathcal{T}_0} I^{\bar{p}}C_{\bullet}^T(X)\right) = H_*(I^{\bar{p}}C_{\bullet}^T(X)) = I^{\bar{p}}H_*^T(X).$$

2 Compute $\mathbb{R}^2$ with two degeneracy lines

In this section, we compute the example in Fig 2. It is $\mathbb{R}^2$ with two degeneracy lines L_1 and L_2 .

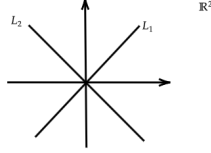


Figure 2: $\mathbb{R}^2$ with two degeneracy lines L_1 and L_2

We have filtration $X_2 \supseteq X_1 \supseteq X_0 \supseteq X_{-1}$ where $X_2 = \mathbb{R}^2$, $X_1 = L_1 \cup L_2$, $X_0 = L_1 \cap L_2$.

The strata are shown in Fig3

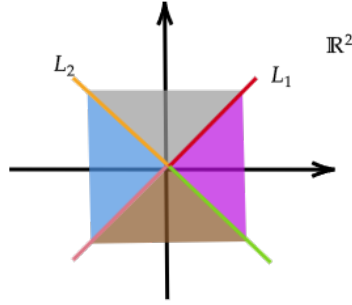


Figure 3: Stratums: There are four strata in $X_2 - X_1$, depicted in blue, brown, purple, grey; There are four strata in $X_1 - X_0$, depicted in red, pink, yellow and green; There is one stratum in X_0 at origin

We only need to choose a full triangulation to compute PL intersection homology. We pick triangulation in Fig4.

The methodology to compute is classifying i -simplexes into several types, see Fig5

We assume perversity are same on O_1, O_2, O_3, O_4 :

$$\bar{p} : \{\text{Strata of } X\} \rightarrow \mathbb{Z}$$

$$R_i \mapsto 0, i = 1, 2, 3, 4$$

$$O_i \mapsto \bar{p}(1), i = 1, 2, 3, 4$$

$$T \mapsto \bar{p}(2)$$

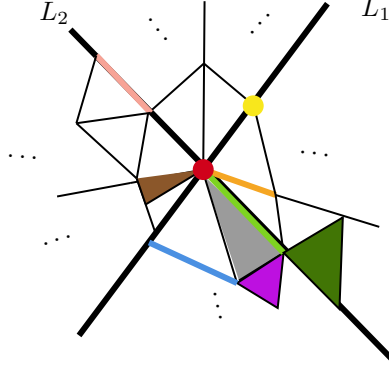


Figure 4: A full triangulation

0-simplex:	type I_0	Intersects with X_0
	type II_0	Intersects with $X_1 - X_0$
1-simplex:	type I_1	Only intersects with $X_1 - X_0$ of dim 1
	type II_1	Only intersects with $X_1 - X_0$ of dim 0
	type III_1	Intersects with $X_1 - X_0, X_0$
	type IV_1	Only intersects with X_0
2-simplex:	type I_2	Intersects with $X_1 - X_0, X_0$
	type II_2	Only intersects with $X_1 - X_0$ of dim 1
	type III_2	Only intersects with $X_1 - X_0$ of dim 0
	type IV_2	Only intersects with X_0

Figure 5: Types of simplexes: we ignore the intersection with $X_2 - X_1$ since we can always ignore the allowability condition on regular strata. The sub-index of a type denotes the dimension of the simplex, and the corresponding examples are depicted in the same color in Fig4

Remark 2.1. The assumption $\bar{p}(O_1) = \bar{p}(O_2) = \bar{p}(O_3) = \bar{p}(O_4)$ for simplicity. We shall throw this assumption after we compute this simple case.

The following steps compute the allowability condition for simplexes of dimension 0, 1, 2.

- Let η be any 2-simplex. It is allowable if
 - $\dim(\eta \cap X_0) \leq 2 - 2 + \bar{p}(2) = \bar{p}(2)$
 - $\dim(\eta \cap (X_1 - X_0)) \leq 2 - 1 + \bar{p}(1) = \bar{p}(1) + 1$
- Let e be any 1-simplex. It is allowable if
 - $\dim(e \cap X_0) \leq 1 - 2 + \bar{p}(2) = \bar{p}(2) - 1$
 - $\dim(e \cap (X_1 - X_0)) \leq 1 - 1 + \bar{p}(1) = \bar{p}(1)$
- Let v be any 0-simplex. It is allowable if
 - $\dim(v \cap X_0) \leq -2 + \bar{p}(2)$
 - $\dim(v \cap (X_1 - X_0)) \leq -1 + \bar{p}(1)$

These allowability conditions suggest that we need to discuss the problem in 16 cases based on the value ranges of $\bar{p}(1)$ and $\bar{p}(2)$. In each of the 16 cases, we have labeled the allowable simplexes for that case; see Fig6.

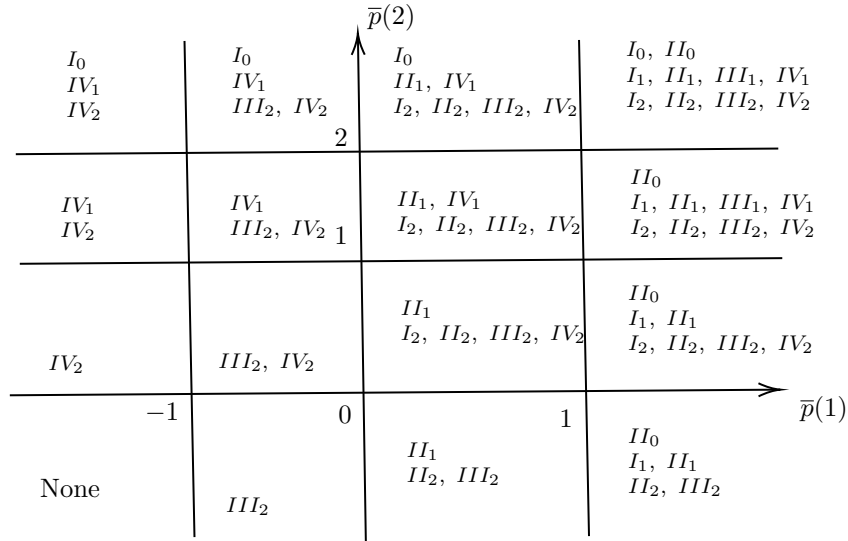
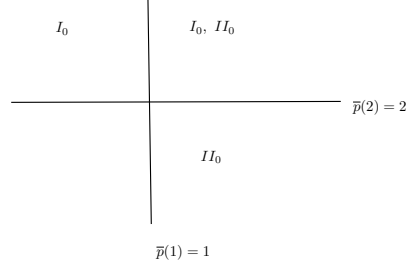
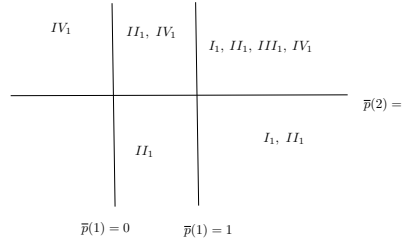


Figure 6: The allowable simplex type in each case. Note that all cases contain points on the line. e.g., the bottom right box is $\bar{p}(2) < 0, \bar{p}(1) \geq 1$.

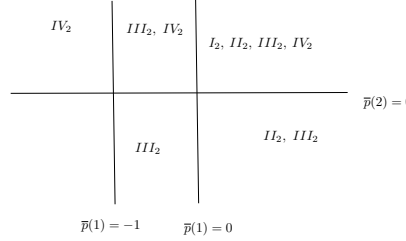
Remark 2.2. Fig6 is obtained by combining three figures in Fig7.



(a) Analysis for 0-simplex



(b) Analysis for 1-simplex



(c) Analysis for 2-simplex

Figure 7: Allowable simplexes

Example 2.3. Consider the case $1 \geq \bar{p}(2) < 2$, $\bar{p}(1) < -1$. In this case, we can see in Fig6 that types IV_1 and IV_2 are allowable. With IV_1 , any allowable 0-simplex can be homologous to each other, so $I^{\bar{p}}H_0 = \mathbb{Z}$. The allowable 1-cycles lie in regular strata, so all 1-cycles are trivial, thus $I^{\bar{p}}H_1 = 0$. There is no 2-cycle, so $I^{\bar{p}}H_2 = 0$.

The other cases are obtained similarly.

The computation results are shown in Fig8.

Example 2.4. We provide some example loops, see Fig9. For $\bar{p}(1) < 0$, $\bar{p}(2) \geq 1$, the loops in (a) are trivial, i.e., they are bounded by allowable 2-chains.

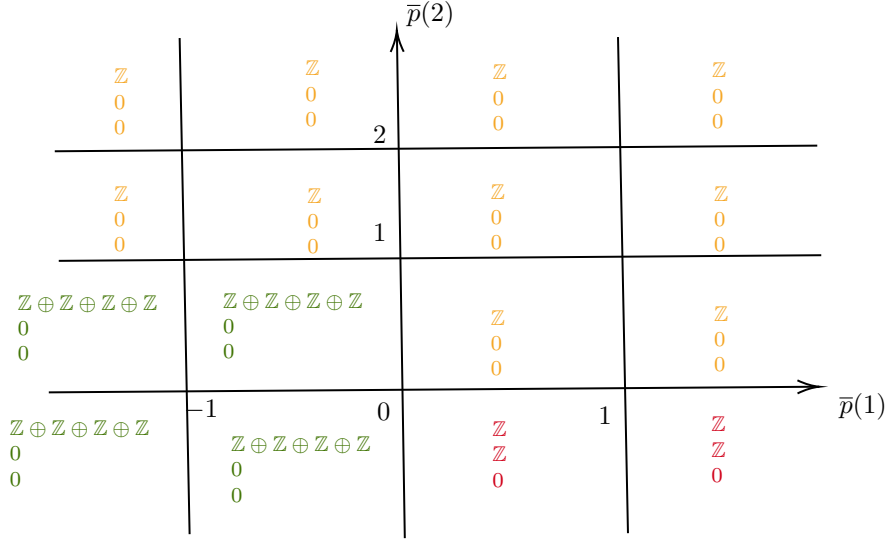


Figure 8: Intersection homology of $\mathbb{R}^2$ with two singular lines: from top to bottom is $I^{\bar{p}}H_0$, $I^{\bar{p}}H_1$, $I^{\bar{p}}H_2$

For $\bar{p}(1) < 0$, $\bar{p}(2) < 1$, the only allowable loops in (b) are trivial loops in regular stratum.

Consider the case $\bar{p}(1) \geq 1$, $\bar{p}(2) \leq 0$ in (c), the loop 1, 2, 3 are trivial and loop 4 is nontrivial, generated $I^{\bar{p}}H_1 = \mathbb{Z}$.

Analysis the results:

- In Fig8, from yellow region to green region, we “see” the two singular lines; From yellow region to red region, we “see” the singular point at the origin.
- It's equal to regular homology $H_*(\mathbb{R}^2)$ for nonnegative perversity. More discussion can be seen in Conjecture 3.4.
- We divide the results in Fig8 into four parts: A, B, C, D, see Fig10.

In part A, type IV_1 are allowable so $I^{\bar{p}}H_0 = \mathbb{Z}$. The only allowable 1-simplex is IV_1 , so any 1-cycle lies in regular strata or touches X_0 . Hence $I^{\bar{p}}H_1 = 0$.

In part B, all 2-simplex are allowable, so $I^{\bar{p}}H_1 = 0$. Since type II_1 is allowable in any case, so $I^{\bar{p}}H_0 = \mathbb{Z}$.

In part C, four types of 1-simplexes are not allowable, so there are four path components, and thus $I^{\bar{p}}H_0 = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$. All allowable 1-cycle lie in regular stratum, so $I^{\bar{p}}H_1 = 0$.

In part D, the type II_1 is allowed, so $I^{\bar{p}}H_0 = \mathbb{Z}$. There is a nontrivial 1-cycle that is the one encloses the origin.

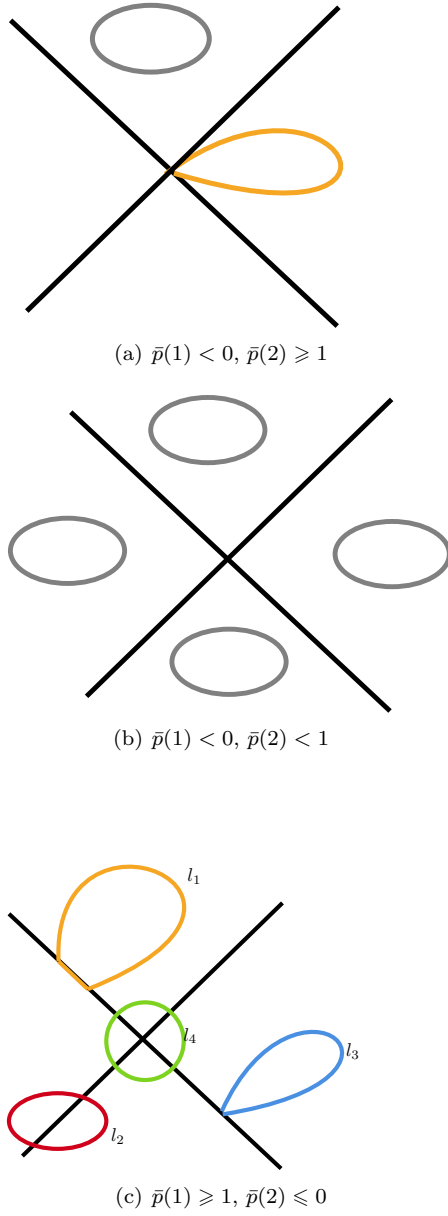


Figure 9: Example loops

Conjecture 2.5. With nonnegative perversity, the intersection homology of smooth 2-manifold is the same as the general homology.

Proof. (Not strict)

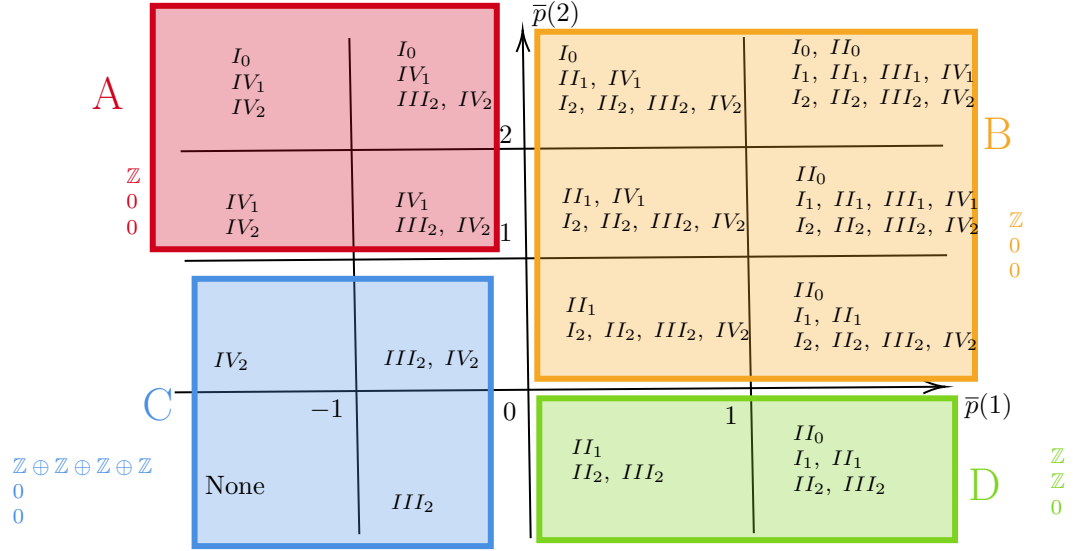


Figure 10: We divide the results into A, B, C, D four parts

- Let η be any 2-simplex. It's allowable if
 - $\dim(\eta \cap X_0) \leq 2 - 2 + \bar{p}(2) = \bar{p}(2)$
 - $\dim(\eta \cap (X_1 - X_0)) \leq 2 - 1 = \bar{p}(1) = \bar{p}(1) + 1$

When perversity is nonnegative, all 2-simplexes are allowable up to homologous, i.e., $C_2^{\bar{p}}(X) = C_2(X)$ up to homologous.

- Let e be any 1-simplex. It is allowable if
 - $\dim(e \cap X_0) \leq 1 - 2 + \bar{p}(2) = \bar{p}(2) - 1$
 - $\dim(e \cap (X_1 - X_0)) \leq 1 - 1 + \bar{p}(1) = \bar{p}(1)$

From the condition, we find that all 1-simplices intersect with $X_1 - X_0$ of 0-dimension, and uncontained X_0 is allowable. Since all 2-simplexes are allowable and it is a smooth manifold, we can always (I guess) homologous a 1-simplex to intersect with $X_1 - X_0$ of 0-dimension and do not contain X_0 . Hence all 1-simplex is allowable up to homologous, i.e., $C_1^{\bar{p}}(X) = C_1(X)$ up to homologous.

- Let v be any 0-simplex. It is allowable if
 - $\dim(v \cap X_0) \leq -2 + \bar{p}(2)$
 - $\dim(v \cap (X_1 - X_0)) \leq -1 + \bar{p}(1)$

We find that all 0-simplex not intersecting X_1 is allowable from the condition. Since all 1-simplexes are allowable and we are in a smooth manifold, we can always (I guess) homologous a 0-simplex to a 0-simplex not intersect X_1 . So, all 0-simplex is allowable up to homologous, $C_0^{\bar{p}}(X) = C_0(X)$ up to homologous.

Hence, $I^{\bar{p}}H_*(X) = H_*(X)$. □

Compute for the perversity that is different in the strata, the interesting result is that when $\bar{p}(2) < 0$, $\bar{p}(O_1) = \bar{p}(O_2) \geq 1$ and $\bar{p}(O_3) = \bar{p}(O_4) < 0$, we have $I^{\bar{p}}H_0(X) = \mathbb{Z} \oplus \mathbb{Z}$, $I^{\bar{p}}H_1(X) = 0$, $I^{\bar{p}}H_2(X) = 0$, which means that we only see one of the two singular lines. Other cases are the same and not very interesting. The non-GM case is computed the same as the GM case but still needs to be rechecked.

3 Compute $\mathbb{R}^2$ with one singular point

The parametrization space of 2-band Hermitian system is $\mathbb{R}^2$ with a single singular at origin, see Fig11.

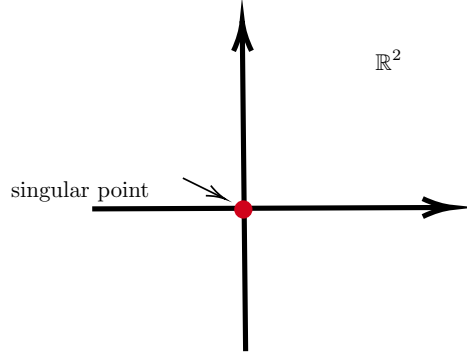


Figure 11: $\mathbb{R}^2$ with a singular point

The types of simplexes are in Fig12 and the allowable simplex in each case are in Fig13.

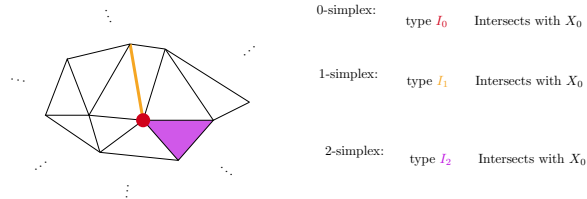


Figure 12: Types of simplexes

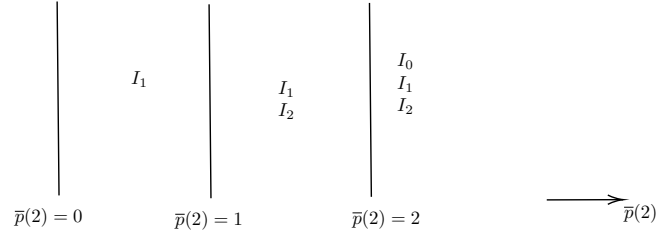


Figure 13: Allowable simplex in each case

By similar calculation, the results are in Fig14.

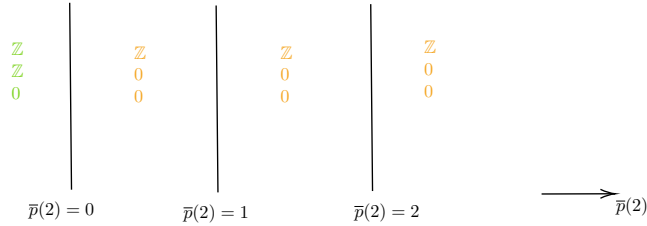


Figure 14: Intersection homology of $\mathbb{R}^2$ with one singular point: from top to bottom is $I^{\bar{p}}H_0$, $I^{\bar{p}}H_1$, $I^{\bar{p}}H_2$

Analysis: From yellow region to green region, the $I^{\bar{p}}H_1(\mathbb{R})$ varies from 0 to $\mathbb{Z}$, meaning that we detect the singular point at the origin.

4 Intersection homology of swallowtail

We consider the filtered space $\mathbb{R}^3$ with filtration $X_3 \supset X_2 \supset X_1 \supset X_0 \supset X_{-1} = \emptyset$ where $X_3 = \mathbb{R}^3$, X_2 is the red surface, X_1 is the three blue intersection lines and X_0 is the origin point, see Fig15.

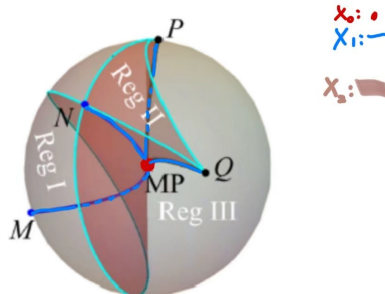


Figure 15: Filtration of swallowtail

Since there is no nontrivial 3-cycle in $C_3(\mathbb{R}^3)$, so is $I^{\bar{p}}C_3(\mathbb{R}^3)$. Thus $I^{\bar{p}}H_3(\mathbb{R}^3) = 0$.

Degree 0 intersection homology

Type of 0-simplex and 1-simplex is shown in Fig16.

0-simplex:	type I_0	Only intersects with X_0
	type II_0	Only intersects with $X_1 - X_0$
	type III_0	Only intersects with $X_2 - X_1$
	type I_1	Only intersects with $X_1 - X_0$ at dim 1
1-simplex:	type II_1	Only intersects with $X_2 - X_1$ at dim 1
	type III_1	Intersects with X_0 at dim 0 and with $X_1 - X_0$ at dim 1
	type IV_1	Intersects with $X_3 - X_2$ at dim 1 and with $X_1 - X_0$ at dim 0
	type V	Intersects with $X_2 - X_1$ at dim 1 and with $X_1 - X_0$ at dim 0
	type VI_1	Intersects with X_0 at dim 0 and with $X_2 - X_1$ at dim 1
	type VII_1	Intersects with $X_3 - X_2$ at dim 1 and with $X_2 - X_1$ at dim 0
	type $VIII_1$	Intersects with X_0 at dim 0 and with $X_3 - X_2$ at dim 1

Figure 16: Types of simplices

The allowable 1 and 0-simplex for different perversity is shown in Fig17.

How to obtain Figure17: Here we take type III_1 for example. Similar argument in Section?, the allowable 1-simplex e satisfies the inequality

$$\begin{cases} \dim(e \cap X_0) \leq \bar{p}(3) - 2 \\ \dim(e \cap (X_1 - X_0)) \leq \bar{p}(2) - 1 \\ \dim(e \cap (X_2 - X_1)) \leq \bar{p}(1) \end{cases}$$

We check Fig16 that the type III_1 intersects X_0 at 0 and intersects $X_1 - X_0$ at dimension 1. Taking the left hand side be 0 and 1 in the first and second equation, respectively, we have the type III_1 is allowable when $0 \leq \bar{p}(3) - 2$ and $1 \leq \bar{p}(2) - 1$, i.e., $\bar{p}(3) \geq 2$ and $\bar{p}(2) \geq 2$, which is shown in Fig17. Also note that we omit the intersection at $X_3 - X_2$ since it's a regular stratum.

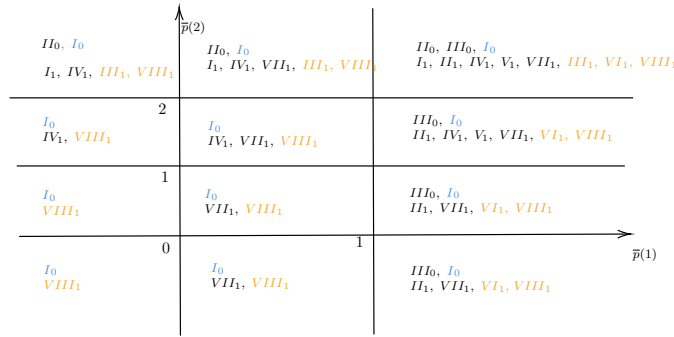


Figure 17: Classes of perversity. The yellow types are allowable when $\bar{p}(3) \geq 2$ and the blue ones are allowable when $\bar{p}(3) \geq 3$.

Sketch of computation: To illustrate more explicitly, we divide the $\bar{p}(1)\bar{p}(2)\bar{p}(3)$ -space into three regions. In the region A, the type IV_1 simplexes are allowable,

which means we can have path across singular lines X_1 to connect three connected components of $X_3 - X_2$; In the region B , the type VII_1 simplexes are allowable, which means we can have path across singular surface $X_2 - X_1$ to connect three connected components of $X_3 - X_2$; In the region C with $\bar{p}(3) \geq 2$, the type $VIII_1$ simplexes are allowable, which means we can have path across X_0 to connects the three connected components of $X_3 - X_2$; In the region C with $\bar{p}(3) < 2$, no simplexes hitting singular part X_2 are allowable, hence we could see the three connected components of $X_3 - X_2$.

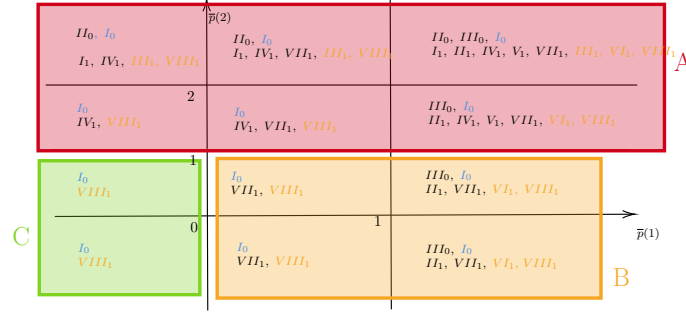


Figure 18: We divided the $\bar{p}(1)\bar{p}(2)\bar{p}(3)$ -space into 3 parts

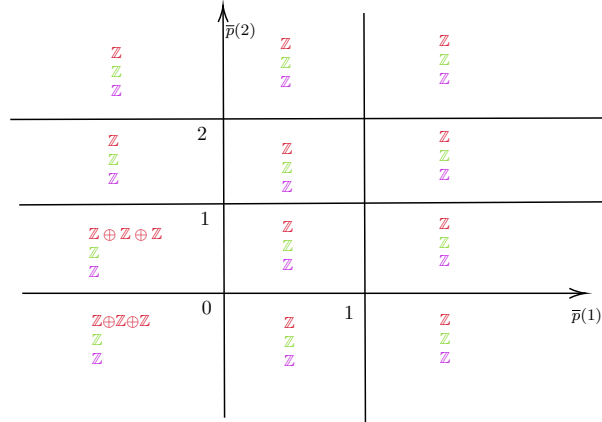


Figure 19: The red is for $\bar{p}(3) < 2$; the green is for $2 \leq \bar{p}(3) < 3$; the purple is for $\bar{p}(3) \geq 3$

The degree 0 intersection homology is shown in Fig19, which can be formulated as:

the degree-0 intersection homology=

$$\begin{cases} \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} & \text{if } \bar{p}(1) < 0, \bar{p}(2) < 1, \text{ and } \bar{p}(3) < 2 \\ \mathbb{Z} & \text{otherwise} \end{cases}$$

Analysis: 0-degree intersection homology depict connected components. By regulating the three parameters, the degree 0 intersection group changes from $\mathbb{Z}$ to $\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$, during which we detect the three regions Reg I, II, III, which are the three connected components of the complementary space of swallowtail in $\mathbb{R}^3$, see Fig15.

Degree 1 intersection homology

Type of 2-simplex is as following:

Table 1: Types of simplices

type I_2	Only intersects with $X_2 - X_1$ at dim 2
type II_2	Intersects with $X_3 - X_2$ at dim 2 and with X_0 at dim 0
type III_2	Intersects with $X_2 - X_1$ at dim 2 and with X_0 at dim 0
type IV_2	Intersects with $X_3 - X_2$ at dim 2 and with $X_1 - X_0$ at dim 1
type V_2	Intersects with $X_3 - X_2$ at dim 2 and with $X_1 - X_0$ at dim 0
type VI_2	Intersects with $X_2 - X_1$ at dim 2 and with $X_1 - X_0$ at dim 1
type VII_2	Intersects with $X_2 - X_1$ at dim 2 and with $X_1 - X_0$ at dim 0
type $VIII_2$	Intersects with $X_3 - X_2$ at dim 2 and with $X_2 - X_1$ at dim 1
type IX_2	Intersects with $X_3 - X_2$ at dim 2 and with $X_2 - X_1$ at dim 0
type X_2	Intersects with $X_2 - X_1$ at dim 2, with $X_1 - X_0$ at dim 1, and with X_0 at dim 0
type XI_2	Intersects with $X_3 - X_2$ at dim 2, with $X_2 - X_1$ at dim 1, and with X_0 at dim 0
type XII_2	Intersects with $X_3 - X_2$ at dim 2, with $X_1 - X_0$ at dim 1, and with X_0 at dim 0
type $XIII_2$	Intersects with $X_3 - X_2$ at dim 2, with $X_2 - X_1$ at dim 1, and with $X_1 - X_0$ at dim 0

The allowable 2 and 1-simplex for different perversity is shown in Fig20.

$I_1, IV_1, III_1, VIII_1$ IV_2, V_2, II_2, XII_2	$I_1, IV_1, III_1, VIII_1$ $IV_2, V_2, IX_2, II_2, XII_2$	$\bar{p}(2) \uparrow$	$I_1, IV_1, VII_1, III_1, VIII_1$ $IV_2, V_2, VIII_2, IX_2, XIII_2,$ II_2, XI_2, XII_2	$I_1, II_1, IV_1, V_1, VII_1, III_1, VI_1, VIII_1$ $I_2, IV_2, V_2, VI_2, VII_2, VIII_2, IX_2, XIII_2,$ $II_2, III_2, X_2, XI_2, XII_2$		
$IV_1, VIII_1$ IV_2, V_2, II_2, XII_2	$IV_1, VIII_1$ $IV_2, V_2, IX_2, II_2, XII_2$	2	$IV_1, VII_1, VIII_1$ $IV_2, V_2, VIII_2, IX_2, XIII_2,$ II_2, XI_2, XII_2	$II_1, IV_1, V_1, VII_1, VI_1, VIII_1$ $I_2, IV_2, V_2, VI_2, VII_2, VIII_2, IX_2, XIII_2,$ $II_2, III_2, X_2, XI_2, XII_2$		
$VIII_1$ V_2, II_2	$VIII_1$ V_2, IX_2, II_2	1	$VII_1, VIII_1$ $V_2, VIII_2, IX_2, XIII_2,$ II_2, XI_2	$II_1, VII_1, VI_1, VIII_1$ $I_2, V_2, VII_2, VIII_2, IX_2, XIII_2,$ II_2, III_2, XI_2		
$VIII_1$ II_2	$VIII_1$ IX_2, II_2	-1	0	1	$II_1, VII_1, VI_1, VIII_1$ $I_2, VIII_2, IX_2,$ II_2, III_2, XI_2	$\bar{p}(1) \rightarrow$

Figure 20: Classes of perversity. The blue is allowable when $\bar{p}(3) \geq 1$ and the yellow is allowable when $\bar{p}(3) \geq 2$

Sketch of computation: Some important loops potentially could be generators of intersection homology groups; see Fig21. Besides those loops, we also

need to consider some loops that hit (not enclose) the singular spaces $X_2 - X_1$, $X_1 - X_0$, or X_0 .

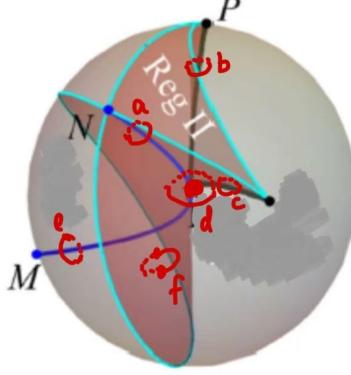


Figure 21: Some important loops

To illustrate more explicitly, we divide the $\bar{p}(1)\bar{p}(2)\bar{p}(3)$ -space into five regions, see Fig 22.

$I_1, IV_1, VII_1, VIII_1$ IV_2, V_2, II_2, XII_2	$I_1, IV_1, VII_1, VIII_1$ $IV_2, V_2, IX_2, II_2, XII_2$	$I_1, IV_1, VII_1, VIII_1$ $IV_2, V_2, VII_2, IX_2, XII_2$ II_2, XI_2, XII_2	$I_1, II_1, IV_1, V_1, VII_1, VIII_1$ $I_2, IV_2, V_2, VII_2, VIII_2, IX_2, XII_2$ $II_2, III_2, X_2, XI_2, XII_2$
IV_1, VII_1 IV_2, V_2, II_2, XII_2	IV_1, VII_1 $IV_2, V_2, IX_2, II_2, XII_2$	$IV_1, VII_1, VIII_1$ $IV_2, V_2, VII_2, IX_2, XII_2$ II_2, XI_2, XII_2	$II_1, IV_1, V_1, VII_1, VIII_1$ $I_2, IV_2, V_2, VII_2, VIII_2, IX_2, XII_2$ $II_2, III_2, X_2, XI_2, XII_2$
$VIII_2$ V_2, II_2	$VIII_2$ V_2, IX_2, II_2	$VII_1, VIII_1$ V_2, VII_2, IX_2, XII_2 II_2, XI_2	$II_1, VII_1, VIII_1, VIII_1$ $I_2, V_2, VII_2, VIII_2, IX_2, XII_2$ II_2, III_2, XI_2
$VIII_1$ II_2	$VIII_1$ IX_2, II_2	$VII_1, VIII_1$ VII_2, IX_2 II_2, XI_2	$II_1, VII_1, VIII_1, VIII_1$ I_2, VII_2, IX_2 II_2, III_2, XI_2

Figure 22: We divided the $\bar{p}(1)\bar{p}(2)\bar{p}(3)$ -space into five parts

In the region A , the allowable 1-cycle cannot transverse the singular surface at $X_2 - X_1$ but can hit X_1 . Since the type V_2 and IV_2 , the 1-cycle e is bounded. Hence, the intersection homology here is all zero.

In the region B , no 1-cycles could hit X_2 . However, the type V_2 is allowable, so the 1-cycle e is trivial. Hence, the intersection homology here is all zero.

In the region C , no 1-cycles are allowed to transverse X_2 . The 1-cycle e is allowed, but the type V_2 is not allowable, making e a generator of intersection homology. Hence, the intersection homology here is all $\mathbb{Z}$.

In the region D , 1-cycles are allowed to transverse the singular surface X_2 . The type V_2 , $VIII_2$, IX_2 , $XIII_2$ are allowed, and the 1-cycles a , b , c , d , e , f are all bounded, so the intersection homology here are all zero.

In the region E with $\bar{p}(3) < 1$, the three 1-cycles a, b, c are all allowed and they are none trivial. Since type $VIII_2$ and IX_2 are allowed, $e = d = a + b + c$ which is also nontrivial. In the region E with $\bar{p}(3) \geq 1$, the type II_2 and XI_2 are allowable, making d (also e) being bounded. Hence the intersection here is $\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}/(a + b + c = 0)$

The degree 1 intersection homology is shown in Fig23, which can be formulated as

The degree-1 intersection homology=

$$\begin{cases} \mathbb{Z}e & \text{if } \bar{p}(1) < 0, \bar{p}(2) < 0, \text{ and } \forall \bar{p}(3) \in \mathbb{Z} \\ \mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c & \text{if } \bar{p}(1) \geq 0, \bar{p}(2) < 0, \text{ and } \bar{p}(3) < 1 \\ \mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c/(a + b + c = 0) & \text{if } \bar{p}(1) \geq 0, \bar{p}(2) < 0, \text{ and } \bar{p}(3) \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

where the generators a, b , and c are shown in Fig??.

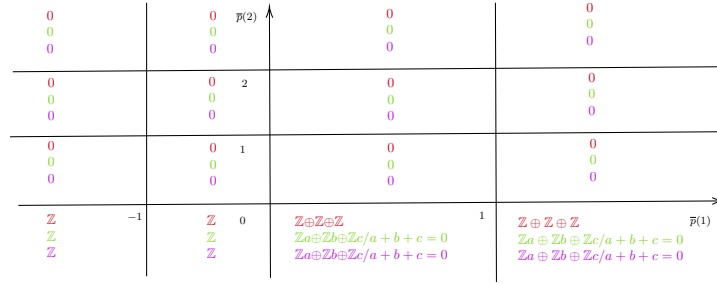


Figure 23: The red is for $\bar{p}(3) < 1$; the green is for $1 \leq \bar{p}(3) < 2$; the purple is for $\bar{p}(3) \geq 2$

Analysis:

- Loops in Fig?? are satisfying $a + b + c = d, d = e$ in intersection groups containing a, b, c, d, e . These equations mean the three singular lines can fuse to the isolated singular lines.
- The group $\mathbb{Z}e$ detects the isolated singular line in the swallowtail but does not detect the three singular lines on the surface.
- The group $\mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c/(a + b + c = 0)$ detects three singular lines on the surface but does not detect the isolated singular line since loop $e = d = 0$.
- The group $\mathbb{Z}a \oplus \mathbb{Z}b \oplus \mathbb{Z}c$ detects both the three singular lines on the surface and the isolated singular line since loop $e = d = a + b + c \neq 0$.

All of the above shows that the isolated singular line strongly relates to the three singular lines on the surface.

Degree 2 intersection homology

The intersection cases of 3-simplex can be classified by intersection dimension. The equivalence condition for those 3-simplexes being allowable is shown in Fig24.

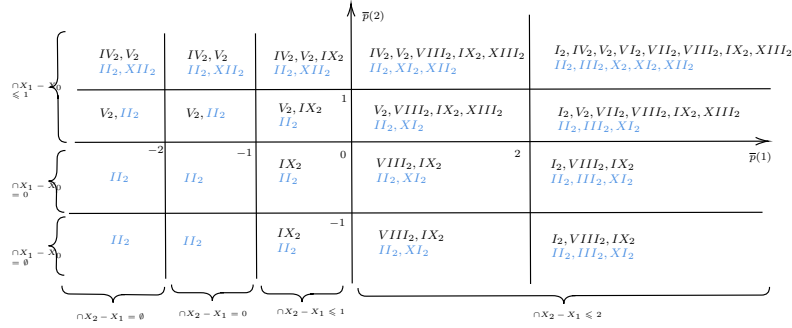


Figure 24: Allowable 2 and 3-simplexes for different perversity conditions; The blue ones are allowable when $\bar{p}(3) \geq 1$; The allowable 3-simplex with different intersection dimension condition are shown in the picture, e.g., the 3-simplex that does not intersect $X_2 - X_1$ is for $\bar{p}(1) < -2$; Note that the condition corresponds to whether 3-simplex could intersect the X_0 is not denoted in the picture. The 3-simplex intersect with X_0 is allowed only when $\bar{p}(3) \geq 0$

Sketch of computation: The possible nontrivial 2-cycles are the ones enclosing the origin, which need the type of 2-simplexes that intersect with $X_1 - X_0$ at ≥ 0 -dimension and $X_2 - X_1$ at ≥ 1 -dimension.

Therefore, for $\bar{p}(3) \geq 0$, the 3-simplex intersecting X_0 is allowed, so the intersection homology for $\bar{p}(3) \geq 0$ are all trivial, i.e., the green and purple one in Fig26 are all zero.

To illustrate more explicitly, we divide the $\bar{p}(1)\bar{p}(2)\bar{p}(3)$ -space into three regions, see Fig25.

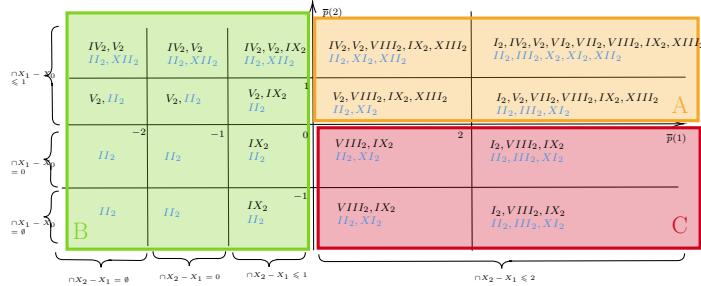


Figure 25: We divided the $\bar{p}(1)\bar{p}(2)\bar{p}(3)$ -space into three parts

In the region A with $\bar{p}(3) < 0$, type V_2 , $VIII_2$, IX_2 are allowable, so the 2-cycle enclose the origin are allowed. However, the 3-simplex cannot intersect

the origin, making this 2-cycle bounded. Hence, the intersection group is $\mathbb{Z}$.

In the region B , no allowable 2-cycles can transverse the singular surface X_2 ; thus, no 2-cycles can enclose the origin. Hence, the intersection homology here is zero.

In the region C , no allowable 2-cycles can hit singular line $X_1 - X_0$, so no 2-cycles can enclose the origin. Hence, the intersection homology here is zero.

The degree 2 intersection homology is shown in Fig26, which can be formulated as

The degree-2 intersection homology=

$$\begin{cases} \mathbb{Z} & \text{if } \bar{p}(1) \geq 0, \bar{p}(2) \geq 0, \text{ and } \bar{p}(3) < 0 \\ 0 & \text{otherwise} \end{cases}$$

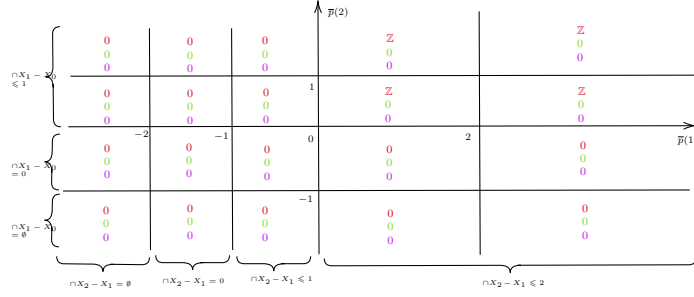


Figure 26: The red is for $\bar{p}(3) < 0$; the green is for $0 \leq \bar{p}(3) < 1$; the purple is for $\bar{p}(3) \geq 1$

Analysis: The $\mathbb{Z}$ group is generated by a sphere enclosing the origin. Hence, during the degree 2 intersection group changing from 0 to $\mathbb{Z}$ by tuning parameters, we detect the singular point MP (meeting point) at the origin.

5 An interesting phenomenon

By the results of intersection homology of $\mathbb{R}^2$ with a singular point at the origin, $\mathbb{R}^2$ with two singular lines, and swallowtail in $\mathbb{R}^3$, we find the following interesting phenomenon: It seems that the degree n intersection homology could only detect the $3 - (n + 1)$ dimension singularities.

Firstly, we need to describe “detect” more explicitly. It is clear that if an n -chain enclose (not intersect with it) a r dimension singularity and no $n + 1$ -chain with n -chain could intersect with r dimension singularity for some perversity, we detect this singularity. For example, a sphere encloses the origin in the swallowtail example. These are all the cases in the above computations.

Since an n -chain encloses an r dimension singularity, which means $r = 3 - (n + 1)$, it seems we describe the phenomenon.

However, are there any other cases? For example, in Fig27, the yellow surface is a singular surface, and the black 1-chain a intersects with these singular surfaces. If there is no allowable 2-chain with 1-chain a being a boundary, then this 1-chain is nontrivial and can detect the singular surface.

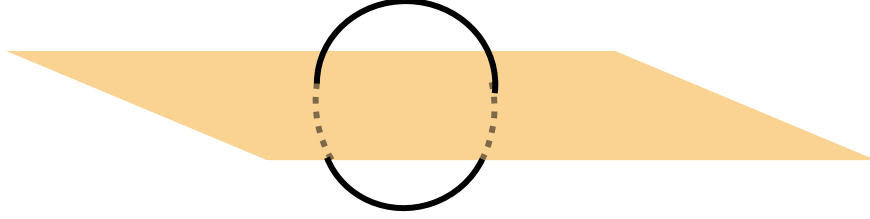


Figure 27: A possible nontrivial chain that not be the enclose type

But it's impossible, by the following property.

Property 5.1. For an allowable n -chain γ , if it intersects with a singular stratum S_i , where S_i has codimension i , and there exists $(n+1)$ -chain η with $\partial\eta = \gamma$ and $\dim(\gamma \cap S_i) \geq \dim(\eta \cap S_i) - 1$, then γ is trivial.

Proof. Consider an allowable n -chain γ and $(n+1)$ -chain η in $C_{n+1}(\mathbb{R}^3)$ with $\partial\eta = \gamma$. η may not be allowable, but we want to show η is allowable, i.e., $\dim(\eta_j \cap S_i) \leq n+1-i+\bar{p}(i)$ for each $(n+1)$ -simplex η_j . Clearly, as a subspace, $\dim(\eta_j \cap S_i) \leq \dim(\eta \cap S_i)$ for each j . Consider the $(n+1)$ -simplex η_{j_0} with $\dim(\eta_{j_0} \cap S_i) = \dim(\eta \cap S_i)$. If η_{j_0} is allowable, then $\dim(\eta_j \cap S_i) \leq \dim(\eta \cap S_i) = \dim(\eta_{j_0} \cap S_i) \leq n+1-i+\bar{p}(i)$ for each j , complete the proof.

Since γ is allowable, so for each n -simplex γ_k in γ , we have $\dim(\gamma_k \cap S_i) \leq n-i+\bar{p}(i)$. As a subspace, we also have $\dim(\gamma_k \cap S_i) \leq \dim(\gamma \cap S_i)$ for each k . Let γ_{k_0} be the n -simplex in γ with $\dim(\gamma_{k_0} \cap S_i) = \dim(\gamma \cap S_i)$. Hence, we have $\dim(\gamma \cap S_i) = \dim(\gamma_{k_0} \cap S_i) \leq n-i+\bar{p}(i)$. Then we obtain that $\bar{p}(i) \geq \dim(\gamma \cap S_i) - n + i$. Since $\dim(\gamma \cap S_i) \geq \dim(\eta \cap S_i) - 1$, $\bar{p}(i) \geq \dim(\eta \cap S_i) - n - 1 + i = \dim(\eta_{j_0} \cap S_i) - n - 1 + i$, meaning that η_{j_0} is allowable.

Hence γ is bounded by η in the intersection homology and thus γ is trivial. $\square$

Example 5.2. The 1-chain γ in Fig27 must be trivial, since $\dim(\gamma \cap S_i) = 0 \geq \dim(\eta \cap S_i) - 1 = 1 - 1 = 0$

Remark 5.3. The condition $\dim(\gamma \cap S_i) \geq \dim(\eta \cap S_i) - 1$ holds for the most cases. Otherwise, $\dim(\gamma \cap S_i) \leq \dim(\eta \cap S_i) - 2$, which is ridiculous since $\dim(\eta) = \dim(\gamma) + 1$.

Note that we discuss the cases where γ intersects with only one strata. It can also be generalized to the case where γ intersects with finitely many strata.

Corollary 5.4. Let Γ be an allowable n -chain intersecting with $\{S_i^1, \dots, S_i^{n_i} | i \in I, n_i \in \mathbb{N}\}$. If there exists an $(n+1)$ -chain η with $\partial\eta = \gamma$ and $\dim(\gamma \cap S_i^l) \leq \dim(\eta \cap S_i^l) - 1$ for all $i \in I$ and $1 \leq l \leq n_i$, then γ is trivial.

Therefore, to have the nonsingular generator of intersection homology, we need the n -chain **NOT** intersects with strata, which means enclosing the singularities.

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- [1] Greg Friedman. *Singular Intersection Homology*. New Mathematical Monographs. Cambridge University Press, 2020.
- [2] Anand Deopurkar. An introduction to intersection homology, 2010. https://deopurkar.github.io/papers/anandrd_minor_thesis.pdf.